

Preimplantation development

Understanding the spatial and temporal developmental processes that take place within an embryo as it develops from a single cell into a recognizable human is the challenge of embryology. The control of these processes resides within the genome; fundamental questions remain concerning the genes and interactions involved in development.

STAGING OF EMBRYOS

For the purposes of embryological study, prenatal life is divided into an embryonic period and a fetal period. The embryonic period covers the first 8 weeks of development (weeks following ovulation and fertilization resulting in pregnancy). The ages of early human embryos have previously been estimated by comparing their development with that of monkey embryos of known postovulatory ages. Because embryos develop at different rates and attain different final weights and sizes, a classification of human embryos into 23 stages occurring during the first 8 weeks after ovulation was developed most successfully by Streeter (1942), and the task was continued by O'Rahilly and Müller (1987). An embryo was initially staged by comparing its development with that of other embryos. On the basis of correlating particular maternal menstrual histories and the known developmental ages of monkey embryos, growth tables were constructed so that the size of an embryo (specifically, the greatest length) could be used to predict its presumed age in postovulatory days (synonymous to postfertilizational days). O'Rahilly and Müller (2000, 2010) emphasize that the stages are based on external and internal morphological criteria and are not founded on length or age. Ultrasonic examination of embryos *in vivo* has necessitated the revision of some of the ages related to stages, and embryos of stages 6–16 are now thought to be up to 3 to 5 days older than the previously used embryological estimates (O'Rahilly and Müller 1999, 2010). Within this staging system, embryonic life commences with fertilization at stage 1; stage 2 encompasses embryos from two cells, through compaction and early segregation, to the appearance of the blastocoele. The developmental processes occurring during the first 10 stages of embryonic life are shown in Figure 8.1.

Much of our knowledge of the early developmental processes is derived from experimental studies on amniote embryos, particularly the chick, mouse and rat. Figure 8.2 shows the comparative timescales of development of these species and human development up to stage 12. The size and age, in postovulatory days, of human development from stage 10 to stage 23 is given in Figure 8.3.

Information on developmental age after stage 23 (8 weeks post ovulation) is shown in Figure 14.3, where the developmental staging used throughout this text is juxtaposed with the obstetric estimation of gestation that is used clinically. A critique of staging terminology and the hazards of the concurrent use of gestational age and embryonic age is given in Chapter 14; sizes and ages of fetuses towards the end of gestation are illustrated in Fig. 14.9.

FERTILIZATION

The central feature of reproduction is the fusion of the two gamete pronuclei at fertilization. In humans, the male gametes are spermatozoa, which are produced from puberty onwards. Female gametes are released as secondary oocytes in the second meiotic metaphase, usually singly, in a cyclical fashion. The signal for the completion of the second meiotic division is fertilization, which stimulates the cell division cycle to resume, completing meiosis and extruding the second polar body (the second set of redundant meiotic chromosomes).

Fertilization normally occurs in the ampullary region of the uterine tube, probably within 24 hours of ovulation. Very few spermatozoa reach the ampulla to achieve fertilization. They must undergo capacitation, a process that is still incompletely understood, and which may

involve modifications of membrane sterols or surface proteins. They traverse the cumulus oophorus and corona radiata, then bind to specific glycoprotein receptors on the zona pellucida, ZP3 and ZP2. Interaction of ZP3 with the sperm head induces the acrosome reaction, in which fusion of membranes on the sperm head releases enzymes, such as acrosin, which help to digest the zona around the sperm head, allowing the sperm to reach the perivitelline space. In the perivitelline space, the spermatozoon fuses with the oocyte microvilli, possibly via two disintegrin peptides in the sperm head and integrin in the oolemma (Figs 8.4, 8.5A).

Fusion of the sperm with the oolemma causes a weak membrane depolarization and leads to a calcium wave, which is triggered by the sperm at the site of fusion and crosses the egg within 5–20 seconds. The calcium wave amplifies the local signal at the site of sperm–oocyte interaction and distributes it throughout the oocyte cytoplasm. The increase in calcium concentration is the signal that causes the oocyte to resume cell division, initiating the completion of meiosis II and setting off the developmental programme that leads to embryogenesis. The pulses of intracellular calcium that occur every few minutes for the first few hours of development also trigger the fusion of cortical granules with the oolemma. The cortical secretory granules release an enzyme that hydrolyses the ZP3 receptor on the zona pellucida and so prevents other sperm from binding and undergoing the acrosome reaction, thus establishing the block to polyspermy. The same cortical granule secretion may also modify the vitelline layer and oolemma, making them less susceptible to sperm–oocyte fusion and providing a further level of polyspermy block.

The sperm head undergoes its protamine → histone transition as the second polar body is extruded. The two pronuclei grow, move together and condense in preparation for syngamy and cleavage after approximately 24 hours (Fig. 8.5B). Nucleolar ribosomal ribonucleic acid (rRNA), and perhaps some messenger RNA (mRNA), are synthesized in pronuclei. A succeeding series of cleavage divisions produces eight even-sized blastomeres at approximately 55 hours, when embryonic mRNA is transcribed.

Several examples of cells that contain male and female pronuclei, termed ootids, have been described. Pronuclear fusion as such does not occur; the two pronuclear envelopes disappear and the two chromosome groups move together to assume positions on the first cleavage spindle. No true zygote containing a membrane-bound nucleus is formed.

The presence of the pronuclei from both parental origins is crucial for spatial organization and the controlled growth of cells, tissues and organs. In the mouse, embryos in which the paternal pronucleus has been removed and replaced with a second maternal pronucleus develop to a relatively advanced state (25 somites), but with limited development of the trophoblast and extraembryonic tissues. In contrast, embryos in which the maternal pronucleus has been replaced by a second paternal pronucleus develop very poorly, forming embryos of only six to eight somites, but with extensive trophoblast. Thus it seems that the maternal genome is relatively more important for the development of the embryo, whereas the paternal genome is essential for the development of the extraembryonic tissues that would lead to placental formation.

This functional inequivalence of homologous parental chromosomes is called parental imprinting. The process causes the expression of particular genes to be dependent on their parental origin; some genes are expressed only from the maternally inherited chromosome and others from the paternally inherited chromosome. The genes involved are called imprinted genes. The requirement for both parental genomes is limited to a subset of the chromosomes. Uniparental disomy can arise through meiotic and mitotic non-disjunction events, and results in individuals who are completely disomic or who exhibit mosaicism of disomic and non-disomic cells. If imprinted genes reside on the affected chromosomes, then the uniparental disomic cells will either express a

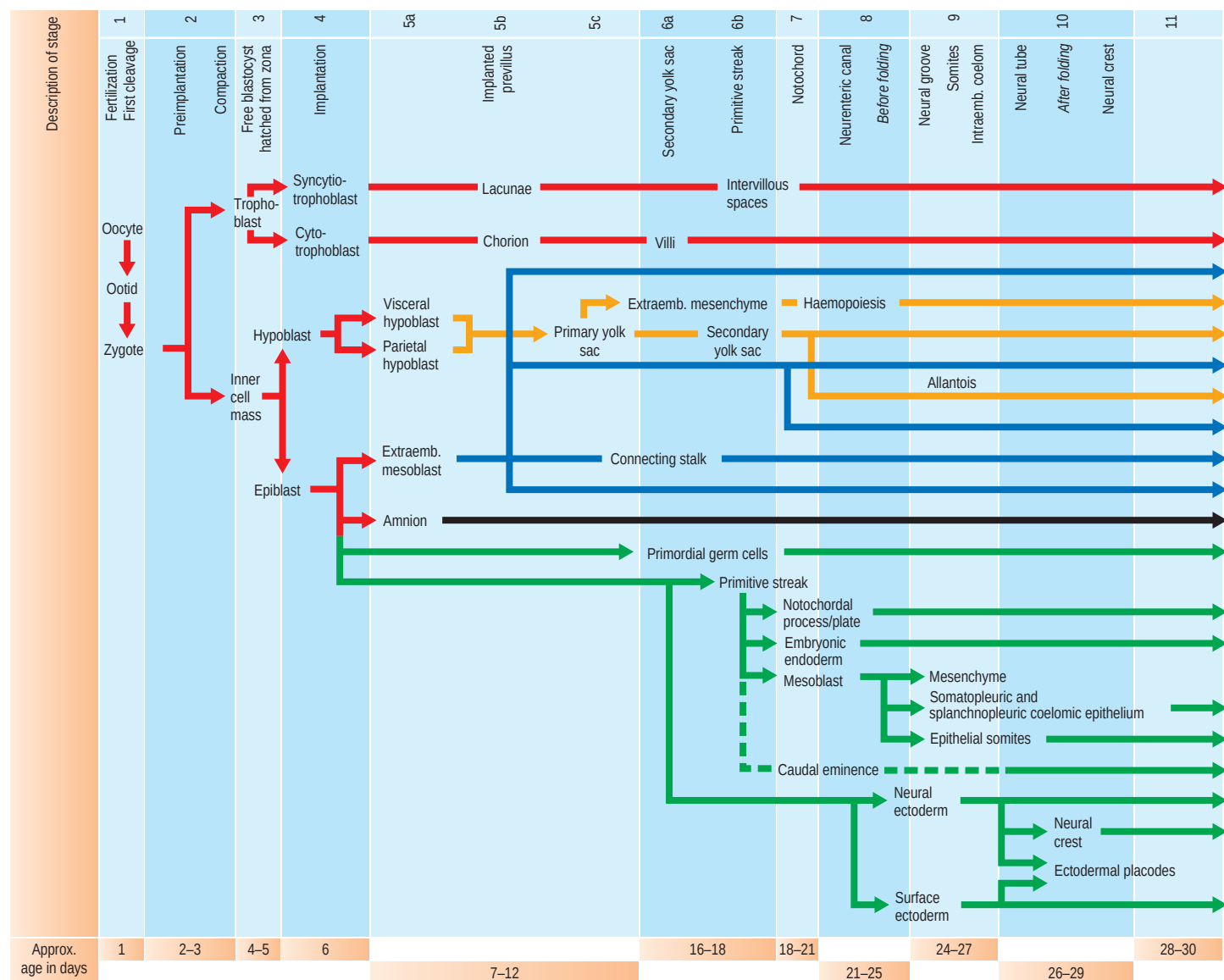


Fig. 8.1 Developmental processes occurring during the first 10 stages of development. In the early stages, a series of binary choices determine the cell lineages. Generally, the earliest stages are concerned with formation of the extraembryonic tissues, whereas the later stages are concerned with the formation of embryonic tissues.

double dose of the gene or have both copies repressed. For example, the gene encoding the embryonic mitogen insulin-like growth factor II is expressed from the paternally inherited chromosome, and repressed when maternally inherited.

In vitro fertilization



In vitro fertilization (IVF) of human gametes is a successful way of overcoming most forms of infertility (Video 8.1). Controlled stimulation of the ovaries (e.g. pituitary downregulation using gonadotrophin-releasing hormone superactive analogues, followed by stimulation with purified follicle stimulating hormone or urinary menopausal gonadotrophins) enables many preovulatory oocytes (often 10 or more) to be recruited and matured, and then aspirated transvaginally using ultrasound guidance, 34–38 hours after injection of human chorionic gonadotrophin (which is given to mimic the luteinizing hormone surge). These oocytes are then incubated overnight with motile spermatozoa in a specially formulated culture medium, in an attempt to achieve successful *in vitro* fertilization. In cases of severe male-factor infertility, in which there are insufficient normal spermatozoa to achieve *in vitro* fertilization, individual spermatozoa can be directly injected into the oocyte in a process known as intracytoplasmic sperm injection (ICSI), which is as successful as routine IVF. In cases in which there are no spermatozoa in the ejaculate, suitable material can sometimes be directly aspirated from the epididymis or surgically retrieved from the testes, and the extracted sperm are then used for intracytoplasmic injection of sperm.

It is also now possible, in some cases, to test embryos for the presence of a particular genetic or chromosomal abnormality by preimplan-

tation genetic diagnosis (PGD). A sample (biopsy) is removed from either the oocyte (polar bodies), the embryo itself (a blastomere) or the blastocyst (small piece of trophoblast), and subjected to specific genetic testing. Oocytes and embryos can also be biopsied and screened for aneuploidy (errors in chromosome copy numbers) in preimplantation genetic screening (PGS). Unaffected embryos can then be identified for transfer to the patient. Embryos that are surplus to immediate therapeutic requirements can also be cryopreserved in liquid nitrogen for later use. Propanediol or dimethylsulphoxide are commonly used as cryoprotectants for early embryos, and, like glycerol, may be used for blastocysts. Conception rates per cycle using ovarian stimulation, IVF and successive transfers of fresh and cryopreserved embryos exceed those obtained during non-assisted, or natural, conception.

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Cleavage

The first divisions of the zygote are termed cleavages. They distribute the cytoplasm approximately equally among daughter blastomeres, so, although the cell number of the preimplantation embryo increases, its total mass actually decreases slightly (Fig. 8.6). Cell division can be asynchronous and daughter cells may retain a cytoplasmic link through much of the immediately subsequent cell cycle via a midbody, as a result of the delayed completion of cytokinesis. No centrioles are present until 16–32 cells are seen, but amorphous pericentriolar material is present and serves to organize the mitotic spindles, which are characteristically more barrel- than spindle-shaped at this time.

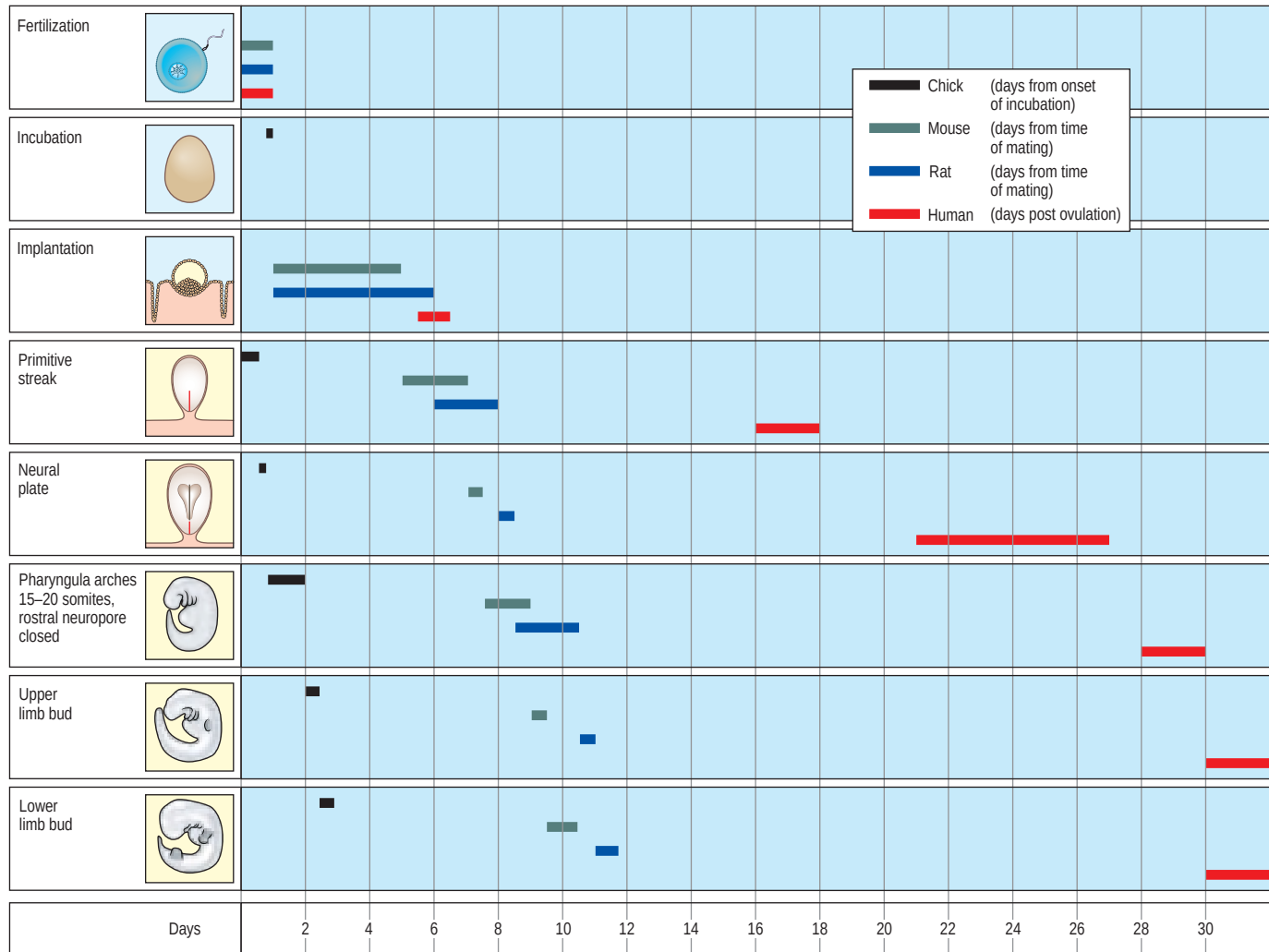


Fig. 8.2 Within developmental biology, evidence concerning the nature of developmental processes has come mainly from studies in vertebrate embryos, most commonly amniote embryos of the chick, mouse and rat. This chart illustrates the comparative timescale of development in these animals and in humans.

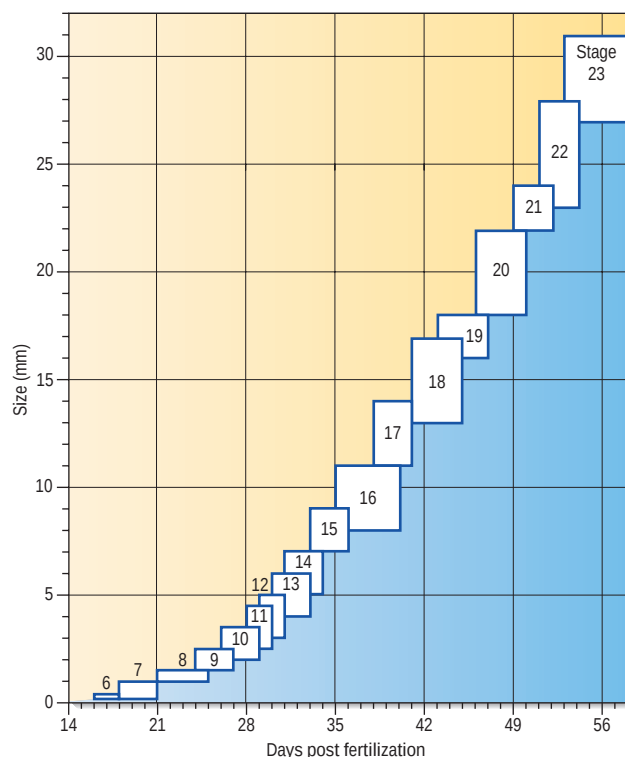


Fig. 8.3 Human developmental stages 10-23. The greatest embryonic length in mm (ordinate) is plotted against age in postfertilizational weeks (abscissa), with the stages superimposed according to current information. (Data provided by courtesy of Professor R O'Rahilly. See also O'Rahilly and Müller (2010).)

All cleavage divisions after fertilization are dependent on continuing protein synthesis. In contrast, passage through the earliest cycles, up to eight cells, is independent of mRNA synthesis. Thereafter, experimental inhibition of transcription blocks further division and development, indicating that activation of the embryonic genome is required. There is also direct evidence for the synthesis of embryonically encoded proteins at this time. As the genes of the embryo first become both active and essential, the previously functional maternally derived mRNA is destroyed. However, protein made on these maternal templates does persist at least during blastocyst growth. Spontaneous developmental arrest of embryos cultured *in vitro* seems to occur during the cell cycle of gene activation, but it is not caused by total failure of that activation process. Early cleavage, up to the formation of eight cells, requires pyruvate or lactate as metabolic substrates, but thereafter more glucose is metabolized and may be required.

The earliest time at which different types of cells can be identified within the cleaving embryo is when 8-16 cells are present. Up to the formation of eight cells, cells are essentially spherical, touch each other loosely, and have no specialized intercellular junctions or significant extracellular matrix; the cytoplasm in each cell is organized in a radially symmetric manner around a centrally located nucleus. Once eight cells have formed, a process of compaction occurs. Cells flatten on each other to maximize intercellular contact, initiate the formation of gap and focal tight junctions, and radically reorganize their cytoplasmic conformation from a radially symmetric to a highly asymmetric phenotype. This latter process includes the migration of nuclei towards the centre of the embryo, the redistribution of surface microvilli and an underlying mesh of microfilaments and microtubules to the exposed surface, and the localization of endosomes beneath the apical cytoskeletal mesh. As a result of the process of compaction, the embryo forms a primitive protoepithelial cyst, which consists of eight polarized cells, in which the apices face outwards and basolateral surfaces face internally. The focal tight junctions, which align to become increasingly linear, are localized to the boundary between the apical and basolateral surfaces.

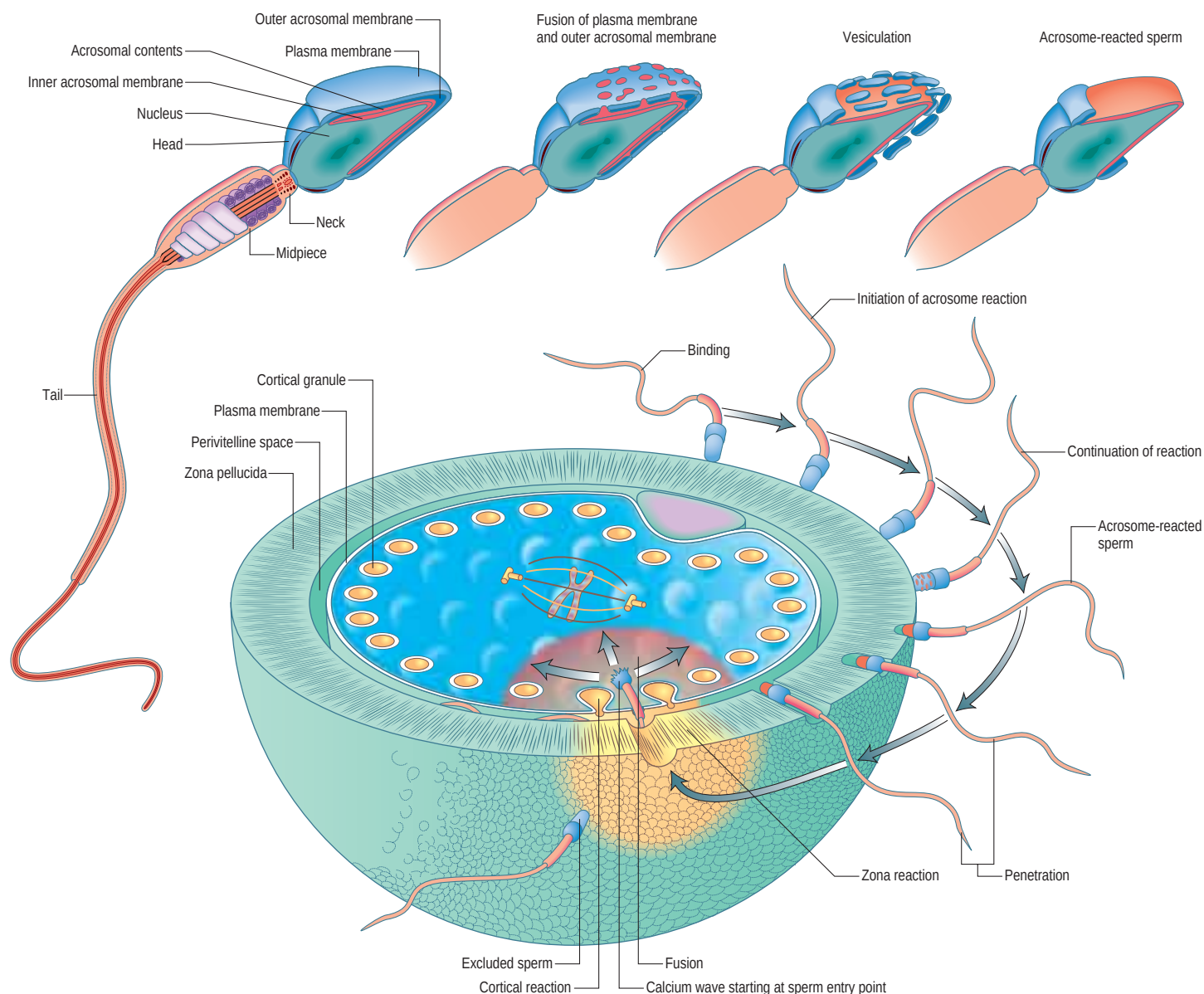


Fig. 8.4 The fertilization pathway: a succession of steps. After a sperm binds to the zona pellucida, the acrosome reaction takes place (see detail at top). The outer acrosomal membrane (blue), an enzyme-rich organelle in the anterior of the sperm head, fuses at many points with the plasma membrane surrounding the sperm head. Then those fused membranes form vesicles, which are eventually sloughed off from the head, exposing the acrosomal enzymes (red). The enzymes digest a path through the zona pellucida, enabling the sperm to advance. Eventually, the sperm meets and fuses with the secondary oocyte plasma membrane and this triggers cortical and zona reactions. First, enzyme-rich cortical granules in the oocyte cytoplasm release their contents (yellow) into the zona pellucida, starting at the point of fusion and progressing right and left. Next, in the zona reaction, the enzymes modify the zona pellucida, transforming it into an impenetrable barrier to sperm as a guard against polyspermy (multiple fertilization).

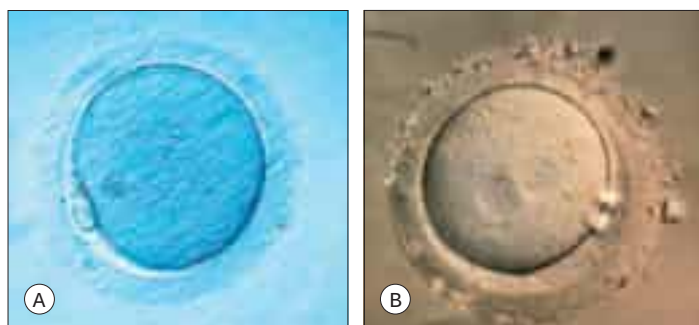


Fig. 8.5 **A**, An unfertilized human secondary oocyte surrounded by the zona pellucida; the first polar body can be seen. Spermatozoa are visible outside the zona pellucida. **B**, A fertilized human ootid before fusion of the pronuclei. Two polar bodies can be seen beneath the zona pellucida.

Gap junctions form between apposed basolateral surfaces and become functional.

The process of compaction involves the cell surface and the calcium dependent cell-cell adhesion glycoprotein, E-cadherin (also called L-CAM or uvomorulin). Neutralization of its function disturbs all three

elements of compaction. The entire process can function in the absence of both mRNA and protein synthesis. Post-translational controls are sufficient and seem to involve regulation through protein phosphorylation. Significantly, although E-cadherin is not synthesized and present on the surface of cleaving blastomeres, it first becomes phosphorylated when eight cells are visible, at the initiation of compaction.

The process of compaction is important for the generation of cell diversity in the early embryo. As each polarized cell divides, it retains significant elements of its polar organization, so that its daughter cells inherit cytoskeletal domains, the nature of which reflects their origin and organization in the original parent cell in the eight-celled embryo. Thus, if the axis of division is aligned approximately at right angles to the axis of cell polarity, the more superficially placed daughter cell inherits all the apical cytoskeleton and some of the basolateral cytoskeleton and is polar, whereas the more centrally placed cell inherits only basolateral cytoskeleton and is apolar. In contrast, if the axis of division is aligned approximately along the axis of the cell polarity, two polar daughter cells are formed. In this way, two-cell populations are formed in the 16-cell embryo that differ in phenotype (polar, apolar) and position (superficial, deep). The number of cells in each population in any one embryo will be determined by the ratio of divisions along, and at right angles to, the axis of eight-cell polarity. The theoretical and observed limits of the polar to apolar ratio are 16:0 and 8:8. The outer polar cells contribute largely to the trophectoderm, whereas the inner

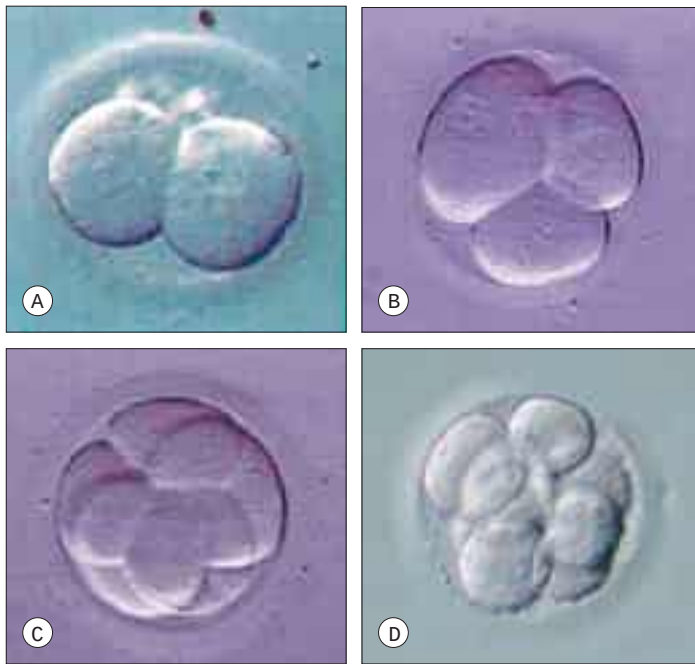


Fig. 8.6 Successive stages of cleavage of a human ootid. **A**, The two-cell stage. **B**, The three-cell stage. **C**, The five-cell stage. **D**, The eight-cell stage.

apolar cells contribute almost exclusively to the inner cell mass in most embryos.

In cleavage, the generation of cell diversity, to either trophectoderm or inner cell mass, occurs in the 16-cell morula and precedes the formation of the blastocyst. During the 16-cell cycle, the outer polar cells continue to differentiate an epithelial phenotype, and display further aspects of polarity and intercellular adhesion typical of epithelial cells, while the inner apolar cells remain symmetrically organized. During the next cell division (16–32 cells), a proportion of polar cells again divide differentially, as in the previous cycle, each yielding one polar and one apolar progeny, which enter the trophectoderm and inner cell mass lineages, respectively. Although differentiative division at this time is less common than at the 8- to 16-cell transition, it has the important function of regulating an appropriate number of cells in the two tissues of the blastocyst. Thus, if differentiative divisions were relatively infrequent at the 8- to 16-cell transition, they will be more frequent at the 16- to 32-cell transition, and *vice versa*.

After division to 32 cells, the outer polar cells complete their differentiation into a functional epithelium, display structurally complete zonular tight junctions and begin to form desmosomes. The nascent trophectoderm engages in vectorial fluid transport in an apical to basal direction to generate a cavity that expands in size during the 32- to 64-cell cycles and converts the ball of cells, the morula, to a sphere, the blastocyst (Fig. 8.7). Once the blastocyst forms, the diversification of the trophectoderm and inner cell mass lineages is complete, and trophectoderm differentiative divisions no longer occur. In the late blastocyst, the trophectoderm is referred to as the trophoblast, which can be divided into polar trophoblast, lying in direct contact with the inner cell mass, and mural trophoblast, surrounding the blastocyst cavity (Fig. 8.8).

Blastocyst

The blastocyst ‘hatches’ from its zona pellucida at 6–7 days, possibly assisted by an enzyme similar to trypsin (see Figs 8.7C, 8.8). Trophoblast oozes out of a small slit; many embryos form a figure-of-eight shape bisected by the zona pellucida, especially if it has been hardened during oocyte maturation and cleavage. Such half-hatching could result in the formation of identical twins. Hatched blastocysts expand and differentiation of the inner cell mass proceeds (see Fig. 8.8).

The outer cells of the blastocyst – the trophoblast or trophectoderm – are flattened polyhedral cells, which possess ultrastructural features typical of a transporting epithelium. The trophoblast covering the inner cell mass is the polar trophoblast and that surrounding the blastocyst cavity is the mural trophoblast. The free, unattached blastocyst is assigned to stage 3 of development at approximately 4 days post ovulation, whereas implantation (before villus development) occurs within a period of 7–12 days post ovulation and over the next two stages of



Fig. 8.7 Human embryos. Formation of a morula and blastocyst within the zona pellucida and blastocyst hatching from the zona pellucida. **A**, A ball of cells, the morula, with the cells undergoing compaction. **B**, The blastocyst cavity is developing and the inner cell mass can be seen on one side of the cavity. **C**, The blastocyst is beginning to hatch from the zona pellucida.



Fig. 8.8 A human blastocyst nearly completely hatched from the zona pellucida. The blastocyst can now expand to its full size.

development. Even at this early stage, cells of the inner cell mass are already arranged into an upper layer (i.e. closest to the polar trophoblast), the epiblast, which will give rise to the embryonic cells, and a lower layer, the hypoblast, which has an extraembryonic fate. Thus, the dorsoventral axis of the developing embryo and a bilaminar arrangement of the inner cell mass are both established at or before implantation. (The earliest primordial germ cells may also be defined at this stage.)

Attachment to the uterine wall

On the sixth postovulatory day, the blastocyst adheres to the uterine mucosa and the events leading to the specialized, intimate contact of trophoblast and endometrium begin. Implantation, which is the term used for this complicated process, includes the following stages: dissolution of the zona pellucida; orientation and adhesion of the blastocyst on to the endometrium; trophoblastic penetration into the endometrium; migration of the blastocyst into the endometrium; and spread and proliferation of the trophoblast, which envelops and specifically disrupts

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and invades the maternal tissues (Ch. 9). The interactions between the hatched blastocyst and the maternal endometrium are not well understood. Developmentally competent embryos appear to enhance the uterine environment and promote their own implantation (Brosens et al 2014, Macklon and Brosens 2014).

The site of implantation is normally in the endometrium of the posterior wall of the uterus, nearer to the fundus than to the cervix; it may be in the median plane or to one or other side, but may occur elsewhere in the uterus, or in an extrauterine or ectopic site.

Ectopic implantation

The conceptus may be arrested at any point during its migration through the uterine tube and implant in its wall. Previous pelvic inflammation damages the tubal epithelium and may predispose to such delay in tubal transport. The presence of an intrauterine contraceptive device or the use of progesterone-based oral contraceptives may also predispose to ectopic pregnancy, probably because of alteration in the normal tubal transport mechanisms.

Nidation of the embryo as an ectopic pregnancy most frequently occurs in the wider ampullary portion of the uterine tube, but may also occur in the narrow intramural part or even in the ovary itself. Most ectopic pregnancies are anembryonic, although the continuing growth of the trophoblast will produce a positive pregnancy test, and may cause rupture of the uterine tube and significant intraperitoneal haemorrhage. Ectopic pregnancies with a live embryo are the most dangerous because they grow rapidly and may be detected only when they have eroded the uterine tube wall and surrounding blood vessels, as early as 8 weeks of pregnancy. Similarly, cornual ectopics (in the intramural part of the tube) may present with catastrophic haemorrhage because there is a substantial blood supply in the surrounding muscularis.

Ovarian or abdominal pregnancies are exceptionally rare. Although some are presumed to have been caused by fertilization occurring in the vicinity of the ovary (primary), most are probably caused secondarily and result from an extrusion of the conceptus through the abdominal ostium of the tube.

Apart from their important clinical implications, these conditions emphasize the fact that the conceptus can implant successfully into tissues other than a normal progestational endometrium. Prolonged development can occur in such sites and is usually terminated by a mechanical or vascular accident and not by a fundamental nutritive or endocrine insufficiency or by an immune maternal response. Abdominal implantation may occur on any organ, e.g. bowel, liver or omentum. If such a pregnancy continues, this makes removal of the placenta at delivery or abortion hazardous as a result of haemorrhage; consequently, the placenta is usually left *in situ* to degenerate spontaneously.

Twinning

Spontaneous twinning occurs once in about every 80 births. Monozygotic twins arise from a single ovum fertilized by a single sperm. At some stage up to the establishment of the axis of the embryonic area and the development of the primitive streak, the embryonic cells separate into two parts, each of which gives rise to a complete embryo. The process of hatching of the blastocyst from the zona pellucida may result in constriction of the emerging cells and separation into two discrete entities. There is a gradual decrease in the average thickness of the zona pellucida with increasing maternal age, which may be causally related to the increase in frequency of monozygotic twinning with increased maternal age. The resultant twins have the same genotype but the description 'identical twins' is best avoided, since most monozygotic twins have differences in phenotypes. Late separation of twins from a single conceptus may result in conjoined twins; these may be equal or unequal, as in acardia. After twinning, monozygotic embryos enter a period of intense catch-up growth. Despite starting out at half the size, each twin embryo or fetus is of a size comparable to a singleton fetus in the second trimester of pregnancy, but declines in relative size in the last 10 weeks of pregnancy. The sex of monozygotic twins will be the same. Monoamniotic, monochorionic, monozygotic twins are most likely to be female, as are acardiac twins. The male to female ratio for all monozygotic twins is 0.487, and for monoamniotic, monochorionic twins it is 0.231.

Dizygotic twins represent the most frequent form of twinning. They result from multiple ovulations, which can be induced by gonadotrophins or drugs commonly used in patients with infertility. Dizygotic twins may be different sexes; like-sex pairs are more common. The male to female ratio is 0.518. Multiple births greater than twinning, such as triplets or quadruplets, can arise from multiple ovulations, or from a

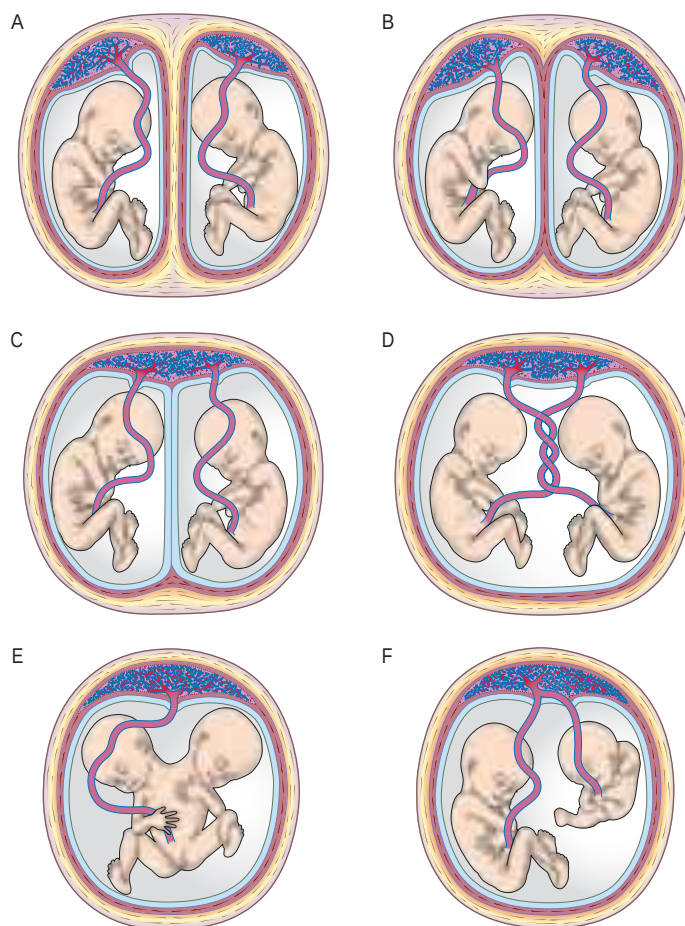


Fig. 8.9 Relationships of the extraembryonic membranes in different types of twinning. **A**, Diamniotic, dichorionic separated; i.e. separation of the first two blastomeres results in separate implantation sites. **B**, Diamniotic, dichorionic fused; here the chorionic membranes are fused but the fetuses occupy separate choria. **C**, Diamniotic, monochorionic; reduplication of the inner cell mass can result in a single placenta and chorionic sacs but separate amniotic cavities. **D**, Monoamniotic, monochorionic; duplication of the embryonic axis results in two embryos sharing a single placenta, chorion and amnion. **E**, Incomplete separation of the embryonic axis results in conjoined twins. **F**, Unequal division of the embryonic axis or unequal division of the blood supply may result in an acardiac monster.

single ovum, or both. It is most likely to be seen in women treated with drugs to stimulate ovulation.

The range of separation of twin embryos is reflected in the separation of the extraembryonic membranes. The types of placentation that can occur are shown in Figure 8.9. Monoamniotic, monochorionic placentae are associated with the greatest perinatal mortality (50%), caused both by entanglement of the umbilical cords impeding the blood supply and by various vascular shunts between the placentae, which may divert blood from one fetus to the other. Artery–artery anastomoses are the most common, followed by artery–vein anastomoses. If the shunting of blood across the placentae from one twin to the other is balanced by more than one vascular connection, development may proceed unimpaired. However, if this is not the case, one twin may receive blood from the other, leading to cardiac enlargement, increased urination and polyhydramnios in the recipient, and anaemia, oligohydramnios and atrophy in the donor.

Dizygotic twins have either completely separate chorionic sacs or sacs that have fused. Such placentae are separated by four membranes, two amnion and two chorion; in addition, these placentae have a ridge of firmer tissue at the base of the dividing membranes, caused by the abutting of two expanding placental tissues against each other.

FORMATION OF EXTRAEMBRYONIC TISSUES

The earliest developmental processes in mammalian embryos involve the production of those extraembryonic structures that will support and nourish the embryo during development. Production of these layers

begins before implantation is complete. At present, it is unclear where the extraembryonic cell lines arise. The trophoblast was considered to be a source but evidence now points to the inner cell mass as the site of origin. **Figure 8.1** shows the sequence of development of various tissues in the early embryo.

Epiblast and amniotic cavity

Epiblast cells are closest to the implanting face of the trophoblast and have a definite polarity; they are arranged in a radial manner with extensive junctions near the centre of the mass of cells, supported by supranuclear organelles. A few epiblast cells are contiguous with cytotrophoblast cells; apart from this contact, a basal lamina surrounds what is initially a spherical cluster of epiblast cells, and isolates them from all other cells. Those epiblast cells adjacent to the hypoblast become taller and more columnar than those adjacent to the trophoblast, and this causes the epiblast sphere to become flattened and the centre of the sphere to be shifted towards the polar trophoblast. Amniotic fluid accumulates at the eccentric centre of the now lenticular epiblast mass, which is bordered by apical junctional complexes and microvilli. As further fluid accumulates, an amniotic cavity forms, roofed by low cuboidal cells that possess irregular microvilli. The cells share short apical junctional complexes and associated desmosomes, and rest on an underlying basal lamina. The demarcation between true amnion cells and those of the remaining definitive epiblast is clear. The columnar epiblast cells are arranged as a pseudostratified layer with microvilli, frequently a single cilium, clefted nuclei and large nucleoli; the cells have a distinct, continuous basal lamina. Cell division in the epiblast tends to occur near the apical surface, causing this region to become more crowded than the basal region. At the margins of the embryonic disc, the amnion cells are contiguous with the epiblast; there is a gradation in cell size from columnar to low cuboidal within a two- to three-cell span (**Fig. 8.10**; see **Fig. 9.1**). Further development of the amnion and amniotic fluid is described on [page 178](#).

Hypoblast and yolk sac

Hypoblast is the term used to delineate the lower layer of cells of the early bilaminar disc, most commonly in avian embryos. This layer is also termed anterior, or distal, visceral endoderm in the mouse embryo. Just before implantation, the hypoblast consists of a layer of squamous cells that is only slightly larger in extent than the epiblast. The cells exhibit polarity, in that apical microvilli face the cavity of the blastocyst and apical junctional complexes, but they lack a basal lamina. During early implantation, the hypoblast extends beyond the edges of the epiblast and can now be subdivided into those cells in contact with the epiblast basal lamina, the visceral hypoblast, and those cells in contact with the mural trophoblast, the parietal hypoblast. The parietal hypoblast cells are squamous; they may share adhesion junctions with the

mural trophoblast and, rarely, may also share gap junctions. The visceral hypoblast cells are cuboidal; they have a uniform apical surface towards the blastocyst cavity but irregular basal and lateral regions, with flanges and projections underlying one another and extending into intercellular spaces. There is no basal lamina subjacent to the visceral hypoblast, and the distance between the hypoblast cells and the epiblast basal lamina is variable.

A series of modifications of the original blastocystic cavity develops beneath the hypoblast later than those developing above the epiblast. While the amniotic cavity is enlarging within the sphere of epiblast cells, the parietal hypoblast cells are proliferating and spreading along the mural trophoblast until they extend most of the way around the circumference of the blastocyst, converging towards the abembryonic pole. At the same time, a space appears between the parietal hypoblast and the mural trophoblast that limits the circumference of the hypoblastic cavity. A variety of terms have been applied to the parietal hypoblast layer: extraembryonic hypoblast, extraembryonic endoderm and exocoelomic (Heuser's) membrane. The cavity that the layer initially surrounds is the primary yolk sac (primary umbilical vesicle), and the resultant smaller cavity lined by hypoblast is the secondary yolk sac. It has been suggested that the secondary yolk sac forms in a variety of ways, including cavitation of visceral hypoblast (a method similar to formation of the amnion), rearrangement of proliferating visceral hypoblast and folding of the parietal layer of the primary yolk sac into the secondary yolk sac. Further development of the yolk sac is described on [page 177](#).

The visceral hypoblast cells are thought to be important in many aspects of the early specification of cell lines and control of the timing of early development. They induce the formation and position of the primitive streak in the midline of the embryo, thus establishing the first axis of the embryonic disc, and they influence the timing of epiblast epithelial-to-mesenchyme transition at the primitive streak. Hypoblast cells remain beneath the primitive streak; their experimental removal in avian embryos causes multiple embryonic axes to form. Hypoblast cells are also believed to be necessary for successful induction of the head region and for the successful specification of the primordial germ cells. With the later formation of the embryonic cell layers from the epiblast, the visceral hypoblast appears to be sequestered into the secondary yolk sac wall by the expansion of the newly formed embryonic endoderm beneath the epiblast, although there is now evidence that some hypoblast cells may remain in the final endoderm layer and contribute to the gut, and that they may also play a role in the induction of cardiogenesis. For a review of the roles of hypoblast cells, see [Stern and Downs \(2012\)](#).

After the formation of the secondary yolk sac, a diverticulum of the visceral hypoblast, the allantois, forms towards one end of the embryonic region and extends into the local extraembryonic mesoblast. It passes from the roof of the secondary yolk sac to the same plane as the amnion. Further development of the allantois is described on [page 178](#).

Extraembryonic mesoblast

By definition, extraembryonic tissues encompass all tissues that do not contribute directly to the future body of the definitive embryo and, later, the fetus. At stage 5, blastocysts are implanted but do not yet display trophoblastic villi (see **Fig. 8.10**); they range from 7 to 12 days in age. A feature of this stage is the first formation of extraembryonic mesoblast, which will come to cover the amnion, secondary yolk sac and the internal wall of the mural trophoblast, and will form the connecting stalk of the embryo with its contained allantoenteric diverticulum. The origin of this first mesoblastic extraembryonic layer is by no means clear; it may arise from several sources, including the caudal region of the epiblast, the parietal hypoblast and subhypoblastic cells. The trophoblastic origin of extraembryonic mesoblast is questioned because there is always a complete basal lamina underlying the trophoblast; the migration of cells out of an epithelium is usually associated with previous disruption of the basal lamina. Certainly, the origin of extraembryonic cells will change over time as new germinal populations are established.

The first mesoblastic extraembryonic layer gives rise to the layer known as extraembryonic mesoblast, arranged as a mesothelium with underlying extraembryonic mesenchymal cells; this also appears to form an extracellular structure corresponding to the magma reticulare, between the mural trophoblast and the primary yolk sac in the stage 5 embryo. Later extraembryonic mesoblast populations mushroom beneath the cytotrophoblastic cells at the embryonic pole, forming the cores of the developing villus stems, and villi and the angioblastic

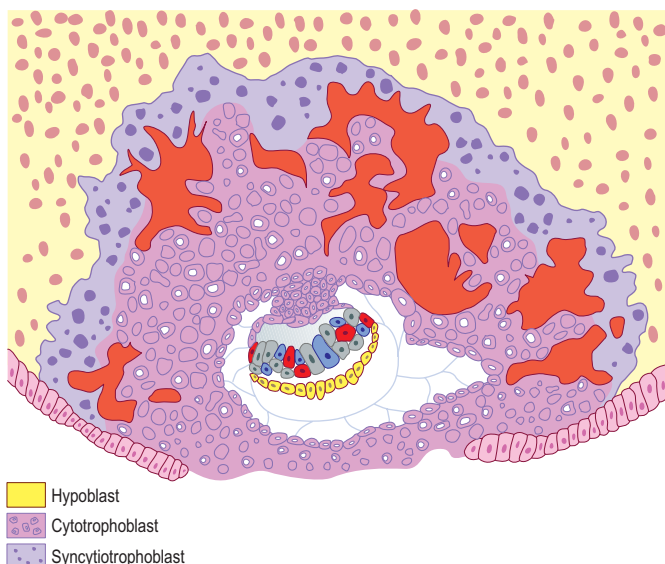


Fig. 8.10 The implanting conceptus at stage 5b. The embryo at this stage is composed only of the abutting epiblast and hypoblast layers. It is suspended within the blastocyst cavity (chorionic cavity) and surrounded by a layer of cytotrophoblast. A large mass of syncytiotrophoblast, with interconnecting lacunae, is penetrating the maternal endometrium.

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cells that will give rise to the capillaries within them and the earliest blood cells.

Initially, the extraembryonic mesoblast connects the amnion to the chorion over a wide area. Continued development and expansion of the extraembryonic coelom means that this attachment becomes increasingly circumvented to a connecting stalk, which is a permanent connection between the future caudal end of the embryonic disc and the chorion. The connecting stalk forms a pathway along which vascular anastomoses around the allantois establish communication with those of the chorion.



Bonus e-book video

Video 8.1 Human *in vitro* fertilization and early development.

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